E-Learning Database Management and Sales Analysis Using MySQL

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1. Project Overview

Developed an E-Learning Database Management System using MySQL to manage learner information, course details, and purchase transactions. The main objective of this project was to organize relational data and perform SQL-based analysis to identify sales trends, course performance, and learner spending patterns.

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2. Database Design

Created a database named `e_learning` with three relational tables:

1. Learners

- learner_id
- full_name
- country

2. Courses

- course_id
- course_name
- category
- unit_price

3. Purchases

- purchase_id
- learner_id
- course_id
- quantity
- purchase_date

Defined appropriate data types and constraints, including Primary Keys and Foreign Keys, to maintain data integrity and establish relationships between the three tables.

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3. Data Insertion

Inserted sample records into all three tables to create a dataset for analysis.

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4. SQL Operations and Analysis

Used various SQL queries and functions to perform data exploration and sales analysis.

- Connected the three tables using INNER JOIN, LEFT JOIN, and RIGHT JOIN.
- Calculated total sales revenue and learner-wise spending.
- Analyzed course category performance.
- Identified the highest and lowest revenue-generating courses.
- Performed year-wise sales analysis.
- Identified the top three most purchased courses based on total quantity sold.

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5. Key Analytical Findings

Overall Sales

- Total Sales Revenue: ₹1,71,000

Category-wise Sales Analysis

Category	Sales
Programming	₹91,500
Web Development	₹12,500

Other categories included Data Analytics and Database.

- Highest Revenue Category: Programming – ₹91,500
- Lowest Revenue Category: Web Development – ₹12,500

Course-wise Revenue Analysis

- Top Revenue-Generating Course: Python – ₹60,000
- Lowest Revenue-Generating Courses: HTML and CSS – ₹12,500

Year-wise Sales Analysis

Year	Sales
2023	₹52,000
2024	₹91,500
2025	₹27,500
Total	**₹1,71,000**

- Highest Sales Year: 2024 – ₹91,500

Learner-wise Spending Analysis

- Highest Spending Learner: Premkumar – ₹59,500 (Coimbatore)
- Lowest Spending Learner: Vishnu Vishal – ₹12,500 (Chennai)

Calculated each learner's total spending based on course price and purchase quantity, along with their country.

Top 3 Most Purchased Courses

Course	Quantity Sold
Power BI	13
Python	12
Java	7

6. Conclusion

This project provided practical experience in relational database management and SQL-based data analysis. By creating and connecting three tables, applying different JOIN operations, and using aggregate functions, I was able to analyze learner spending, course performance, category-wise revenue, and yearly sales trends.

The analysis revealed that Programming generated the highest category revenue, Python was the top revenue-generating course, and 2024 recorded the highest annual sales.

Overall, this project improved my understanding of MySQL database design, relational data management, and analytical problem-solving skills. It also helped me understand how SQL can be used to transform raw data into meaningful business insights for better decision-making.